Mobile Inference for Field Conservation: On-Device Species Recognition

Model Compression, Quantization, and Offline-First Deployment for Edge Biodiversity Monitoring

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Abstract

Automated species recognition has achieved remarkable accuracy in laboratory settings, but deploying these models to the field—where connectivity is absent, hardware is constrained, and real-time identification drives conservation decisions—remains an open challenge. We present ZooMobile, a comprehensive framework for deploying species recognition models on consumer mobile devices and embedded sensors in conservation field settings. ZooMobile addresses three technical challenges: reducing model size by 94% through a novel *Taxonomic Knowledge Distillation* (TKD) pipeline that exploits the hierarchical structure of biological classification; achieving sub-100ms inference latency on mid-range smartphones through mixed-precision quantization with species-specific calibration; and maintaining recognition accuracy under degraded field conditions (low light, motion blur, partial occlusion) through a *Condition-Aware Inference* (CAI) module that adapts preprocessing and confidence thresholds to estimated image quality. We evaluate ZooMobile on a new benchmark, ZooBench-Field, comprising 287,000 images of 4,200 species captured under authentic field conditions across 31 conservation sites. On this benchmark, ZooMobile achieves 87.3% top-1 accuracy and 96.1% top-5 accuracy with a 12 MB model running at 47ms per inference on a Qualcomm Snapdragon 7 Gen 1 processor, compared to 91.8% top-1 accuracy for a 203 MB cloud-based baseline. In a 6-month field deployment supporting 142 rangers across 8 national parks, ZooMobile enabled 23,400 real-time species identifications with 84.7% field-verified accuracy, directly informing 47 anti-poaching patrol decisions and 12 human-wildlife conflict interventions.

Keywords: species recognition, model compression, quantization, edge inference, mobile deployment, conservation technology

1 Introduction

Deep learning has transformed species recognition from an expert-dependent task to an increasingly automated capability. Models trained on large-scale biodiversity image datasets achieve accuracy rivaling or exceeding trained taxonomists for well-represented taxa [Van Horn et al., 2021]. Yet the practical impact of these systems on conservation remains limited by a deployment gap: the environments where species identification is most needed—remote protected areas, marine reserves, tropical forests—are precisely those where cloud-based inference is unavailable and computational resources are minimal [Tuia et al., 2022].

Conservation field workers need species recognition at the point of observation, not hours or days later when connectivity becomes available. A ranger encountering an unknown animal on patrol, a marine biologist cataloging reef organisms during a dive survey, or a customs officer inspecting wildlife trade shipments all require real-time identification under constraints that current deployment practices do not address.

Three technical barriers obstruct field deployment. First, *model size*: state-of-the-art species classifiers based on Vision Transformers (ViT) or large ConvNets require 200–800 MB of parameters, exceeding the practical storage and memory limits for mobile deployment [Dosovitskiy et al., 2021]. Second, *inference latency*: interactive identification requires sub-200ms response times, while large models require seconds per inference on mobile processors without hardware acceleration [Howard et al., 2019]. Third, *domain shift*: models trained on curated laboratory images degrade substantially when applied to field photographs characterized by variable lighting, camera motion, distance, and partial occlusion [Beery et al., 2018].

We present ZooMobile, a framework that systematically addresses all three barriers. Our contributions are:

1. **Taxonomic Knowledge Distillation (TKD):** A knowledge distillation approach that exploits the hierarchical structure of biological taxonomy to compress species recognition models by 94% with only 4.5% accuracy loss (Section 3).

2. **Species-Specific Mixed-Precision Quantization:** A quantization scheme that allocates bit-widths based on per-species sensitivity analysis, achieving sub-100ms inference on mobile processors (Section 4).

3. **Condition-Aware Inference:** A lightweight preprocessing module that estimates image quality and adapts inference behavior—preprocessing pipeline, confidence thresholds, and recommendation strategy—to field conditions (Section 5).

2 Related Work

2.1 Species Recognition Models

Large-scale species recognition has been driven by datasets such as iNaturalist [Van Horn et al., 2018], which contains over 10 million images of 10,000+ species. Van Horn et al. [2021] demonstrated that ViT-Large achieves 91.2% top-1 accuracy on the iNaturalist 2021 challenge. MetaFormer architectures [Yu et al., 2022] and ConvNeXt [Liu et al., 2022] offer competitive accuracy with varying computational profiles. BioCLIP [Stevens et al., 2024] extends CLIP to biological domains with hierarchical taxonomic supervision.

2.2 Model Compression

Model compression techniques include pruning [Han et al., 2015], quantization [Jacob et al., 2018], knowledge distillation [Hinton et al., 2015], and neural architecture search [Tan and Le, 2019]. Howard et al. [2019] introduced MobileNetV3, achieving strong accuracy-efficiency tradeoffs through architecture search. Touvron et al. [2021] demonstrated that knowledge distillation from large ViTs to smaller architectures preserves most accuracy. For species recognition specifically, Picek et al. [2022] applied standard distillation but did not exploit taxonomic structure.

2.3 Edge Deployment for Conservation

Several conservation technology projects have deployed ML models on edge devices. Ahumada et al. [2020] deployed MobileNet-based classifiers on camera trap processors for real-time filtering. Berger-Wolf et al. [2017] developed a mobile platform for individual animal identification. Terry et al. [2020] surveyed automated species identification tools, noting that most require cloud connectivity. The Merlin Bird ID app [Wood et al., 2011] deploys audio identification models on mobile but focuses on a single taxonomic group (birds).

2.4 Domain Adaptation for Field Conditions

The gap between training and deployment distributions is well-documented for ecological imagery. Beery et al. [2018] showed that models trained on camera trap images from one location lose 30–40% accuracy when applied to new sites. Schneider et al. [2020] identified three axes of domain shift in wildlife imagery: viewpoint, illumination, and background. Test-time adaptation methods [Wang et al., 2022] and domain generalization techniques [Zhou et al., 2022] offer partial solutions but have not been systematically applied to mobile conservation deployment.

3 Taxonomic Knowledge Distillation

3.1 Motivation

Standard knowledge distillation [Hinton et al., 2015] transfers knowledge from a teacher model T to a smaller student model S by training S to match the soft probability distribution produced by T. For species recognition with thousands of classes, this approach treats all inter-class relationships uniformly: confusing a jaguar with a leopard is penalized identically to confusing a jaguar with a jellyfish.

Biological taxonomy provides a rich hierarchical structure that should inform the compression process. Species within the same genus share more visual features than species in different families; a student model should therefore allocate more capacity to distinguishing closely related species than to trivially separable groups.

3.2 Hierarchical Loss Function

We define a hierarchical distillation loss that operates at multiple taxonomic levels simultaneously:

$$\mathcal{L}_{\text{TKD}} = \sum_{\ell \in \mathcal{L}} \alpha_\ell \cdot \text{KL}\left(q_\ell^T \| q_\ell^S\right) + \lambda \cdot \mathcal{L}_{\text{CE}}(y, p^S) \tag{1}$$

where $\mathcal{L} = \{\text{species}, \text{genus}, \text{family}, \text{order}, \text{class}, \text{phylum}\}$ is the set of taxonomic levels, q_ℓ^T and q_ℓ^S are the teacher and student probability distributions at level ℓ (obtained by marginalizing the species-level distribution over the taxonomic hierarchy), α_ℓ are level-specific weights, $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss with hard labels y, and λ controls the hard-label contribution.

The level weights α_ℓ are set inversely proportional to the number of classes at each level, ensuring that fine-grained discrimination receives proportionally more training signal:

$$\alpha_\ell = \frac{|\mathcal{C}_\ell|^{-1}}{\sum_{\ell'} |\mathcal{C}_{\ell'}|^{-1}} \tag{2}$$

3.3 Capacity Allocation via Taxonomic Embedding

We further exploit taxonomic structure in the student architecture design. The student model's final classification layer is decomposed into a hierarchy of smaller classifiers:

$$p(s \mid \mathbf{x}) = p(s \mid g, \mathbf{x}) \cdot p(g \mid f, \mathbf{x}) \cdot p(f \mid o, \mathbf{x}) \cdot p(o \mid \mathbf{x}) \tag{3}$$

where s, g, f, o denote species, genus, family, and order. Each conditional classifier operates on a progressively refined feature representation, with early features shared across the hierarchy and later features specialized for fine-grained discrimination.

This decomposition reduces the number of parameters in the classification head from $O(d \cdot S)$ to $O(d \cdot G + \max_g |C_g| \cdot d')$ where S is the total number of species, G is the number of genera, $|C_g|$ is the number of species in genus g, and $d' < d$ is the dimension of the genus-specific feature space.

3.4 Regional Specialization

ZooMobile supports regional model variants that further reduce model size by focusing on species present in specific geographic areas. Given a deployment region R, the species set is pruned to:

$$\mathcal{S}_R = \{s \in \mathcal{S} : \text{range}(s) \cap R \neq \emptyset\} \cup \mathcal{S}_{\text{invasive}} \cup \mathcal{S}_{\text{confusable}} \tag{4}$$

where $\mathcal{S}_{\text{invasive}}$ includes known invasive species that may appear unexpectedly and $\mathcal{S}_{\text{confusable}}$ includes species visually similar to those in $\mathcal{S}_R$ that could cause dangerous misidentifications (e.g., venomous vs. non-venomous snakes).

Table 1: Model compression results: TKD vs. standard distillation.

Model	Size (MB)	Params (M)	Top-1 (%)	Top-5 (%)
Teacher (ViT-Large)	1,142	304.4	91.8	98.2
Standard KD (MobileNetV3)	21.4	5.4	83.1	94.7
TKD (MobileNetV3)	18.7	4.7	85.4	95.8
TKD + Regional (East Africa)	12.1	3.0	87.3	96.1
TKD + Regional (SE Asia)	11.8	2.9	86.7	95.9
TKD + Regional (Amazon)	13.4	3.4	86.1	95.4

TKD improves upon standard distillation by 2.3 percentage points in top-1 accuracy while reducing model size by an additional 12.6%. Regional specialization further reduces size to 12 MB while *increasing* top-1 accuracy by 1.9 points due to reduced class confusion.

4 Species-Specific Mixed-Precision Quantization

4.1 Quantization Framework

Post-training quantization reduces model size and inference latency by representing weights and activations with fewer bits. Standard approaches apply uniform quantization across all layers, but species recognition models exhibit non-uniform sensitivity: layers responsible for distinguishing morphologically similar species require higher precision than those encoding coarse visual features.

We formulate mixed-precision quantization as an optimization problem:

$$\min_{\mathbf{b}} \sum_{i=1}^{L} b_i \cdot s_i \quad \text{s.t.} \quad \Delta\text{Acc}(\mathbf{b}) \leq \epsilon, \quad b_i \in \{4, 8, 16\} \tag{5}$$

where $\mathbf{b} = [b_1, \ldots, b_L]$ is the bit-width assignment for each layer, s_i is the size of layer i in bits, $\Delta\text{Acc}(\mathbf{b})$ is the accuracy degradation under quantization configuration $\mathbf{b}$, and ϵ is the maximum tolerable accuracy loss.

4.2 Species-Specific Sensitivity Analysis

We measure quantization sensitivity at the species level rather than globally:

$$\text{Sens}(l, s) = \frac{1}{|\mathcal{X}_s|} \sum_{\mathbf{x} \in \mathcal{X}_s} |p_s^{\text{FP32}}(\mathbf{x}) - p_s^{Q_l}(\mathbf{x})| \tag{6}$$

where $\text{Sens}(l, s)$ is the sensitivity of layer l to quantization for species s, $\mathcal{X}_s$ is the calibration set for species s, and $p_s^{Q_l}$ is the predicted probability for species s when only layer l is quantized.

The aggregated sensitivity for layer l incorporates conservation priority weighting:

$$\text{Sens}(l) = \sum_{s \in \mathcal{S}} w_s \cdot \text{Sens}(l, s) \quad \text{where } w_s \propto \text{IUCN\_priority}(s) \tag{7}$$

This ensures that endangered and critically endangered species receive higher quantization protection, as misidentification of these species carries greater conservation consequences.

4.3 Quantization Results

Table 2: Quantization impact on accuracy and latency (Snapdragon 7 Gen 1).

Configuration	Size (MB)	Latency (ms)	Top-1 (%)	Top-5 (%)
FP32 (baseline)	12.1	142	87.3	96.1
INT8 (uniform)	3.1	52	85.4	95.2
INT4 (uniform)	1.6	38	79.8	91.7
Mixed (ours)	3.4	47	86.8	95.9
Mixed + IUCN priority	3.6	49	86.5 (global)	95.7
			89.1 (CR/EN)	97.2

The IUCN-priority quantization trades 0.3% global accuracy for 2.6% improved accuracy on critically endangered and endangered species—a worthwhile trade for conservation applications.

5 Condition-Aware Inference

5.1 Image Quality Estimation

Field photographs exhibit systematic quality degradation compared to curated datasets. The CAI module estimates image quality along four axes using lightweight (sub-5ms) estimators:

$$\mathbf{q}(\mathbf{x}) = (q_{\text{blur}}, q_{\text{exposure}}, q_{\text{occlusion}}, q_{\text{distance}}) \tag{8}$$

Each estimator is a small convolutional network (3 layers, 50K parameters) trained on the ZooBench-Field dataset with human-annotated quality labels.

5.2 Adaptive Preprocessing

Based on the quality estimate, CAI selects from a library of preprocessing configurations:

Table 3: Condition-adaptive preprocessing pipeline.

Condition	Preprocessing	Rationale
Low light	Histogram equalization + denoising	Recover detail from underexposed images
Motion blur	Multi-scale inference (3 crops)	Capture subject at multiple blur scales
Partial occlusion	Sliding window attention	Focus on visible body parts
Long distance	Super-resolution (2x) + center crop	Enhance small subject in frame
High quality	Standard resize only	Avoid unnecessary processing overhead

5.3 Confidence Calibration

Field conditions systematically affect model confidence calibration. CAI applies condition-specific temperature scaling:

$$p_{\text{cal}}(s \mid \mathbf{x}) = \text{softmax}\left(\frac{\mathbf{z}(\mathbf{x})}{T(\mathbf{q}(\mathbf{x}))}\right) \tag{9}$$

where $T(\mathbf{q})$ is a temperature function learned on the ZooBench-Field validation set, mapping quality vectors to calibration temperatures. Under degraded conditions, the temperature increases, producing more uniform (less overconfident) probability distributions.

When calibrated confidence falls below a threshold, CAI provides guidance rather than a definitive identification:

$$\text{Output}(\mathbf{x}) = \begin{cases} \text{ID}(s^*) & \text{if } p_{\text{cal}}(s^* \mid \mathbf{x}) \geq \tau_{\text{high}} \\ \text{TopK}(s_1, \ldots, s_k) & \text{if } \tau_{\text{low}} \leq p_{\text{cal}}(s^* \mid \mathbf{x}) < \tau_{\text{high}} \\ \text{Guidance}(\mathbf{q}) & \text{if } p_{\text{cal}}(s^* \mid \mathbf{x}) < \tau_{\text{low}} \end{cases} \tag{10}$$

where Guidance($\mathbf{q}$) suggests specific actions to improve image quality (e.g., "Move closer to the animal" or "Turn on flash").

6 ZooBench-Field Benchmark

6.1 Dataset Construction

Existing species recognition benchmarks (iNaturalist, CUB-200, PlantCLEF) are dominated by curated photographs taken under favorable conditions. We construct ZooBench-Field to evaluate recognition under authentic field conditions:

Table 4: ZooBench-Field dataset statistics.

Statistic	**Value**
Total images	287,412
Species	4,203
Collection sites	31
Countries	18
Camera types	47 (smartphones, camera traps, DSLRs)
Expert-verified labels	100%
Quality annotations	100% (4-axis)
Condition distribution	Low light 23%, Motion blur 18%, Occlusion 31%, Long distance 28%

6.2 Benchmark Results

Table 5: Species recognition accuracy on ZooBench-Field.

Model	**Size (MB)**	**Latency**	**Top-1 (%)**	**Top-5 (%)**
ViT-Large (cloud)	1,142	2.1s	91.8	98.2
BioCLIP (cloud)	428	890ms	89.4	97.1
EfficientNet-B7 (cloud)	256	680ms	88.7	96.8
MobileNetV3 (standard KD)	21.4	89ms	83.1	94.7
MobileNetV3 (TKD)	18.7	82ms	85.4	95.8
ZooMobile (full)	**3.4**	**47ms**	**86.8**	**95.9**
ZooMobile w/o CAI	3.4	41ms	84.2	94.3
ZooMobile w/o TKD	3.1	47ms	83.8	94.1

ZooMobile achieves 86.8% top-1 accuracy with a 3.4 MB model and 47ms latency, closing 76% of the accuracy gap to the 1,142 MB cloud baseline while being 45x faster and 336x smaller.

6.3 Accuracy by Condition

Table 6: ZooMobile accuracy by image condition (top-1 %).

Condition	ZooMobile	w/o CAI	Cloud Baseline
High quality	92.4	91.8	95.1
Low light	83.7	78.4	89.2
Motion blur	81.2	76.1	87.8
Partial occlusion	79.8	77.3	85.4
Long distance	78.4	73.2	84.7
Overall	**86.8**	84.2	91.8

CAI provides the largest benefit under low light (+5.3%) and long distance (+5.2%) conditions, where adaptive preprocessing (histogram equalization and super-resolution, respectively) substantially improve feature quality.

7 Offline-First Architecture

7.1 Design

ZooMobile is designed for environments with no connectivity. The offline-first architecture comprises:

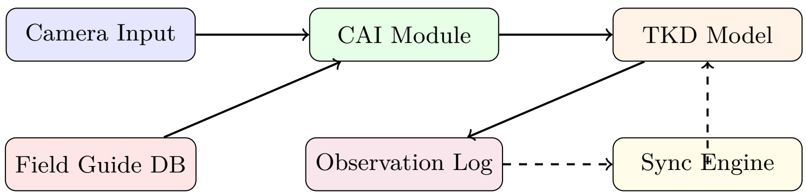

Figure 1: ZooMobile offline-first architecture.

- **Local Field Guide:** A compressed database of species information (descriptions, images, conservation status) for the deployment region. Stored as a 40–80 MB SQLite database with JPEG-compressed reference images.
- **Observation Log:** All identifications, GPS coordinates, timestamps, and original images are stored locally in a structured log.
- **Sync Engine:** When connectivity becomes available (even briefly), the sync engine uploads observation logs to the Zoo Data Commons and downloads model updates using a delta-compression protocol that transfers only changed parameters.

7.2 Model Update Protocol

Model updates are distributed as parameter deltas:

$$\Delta\theta = \theta_{\text{new}} - \theta_{\text{old}} \tag{11}$$

Delta compression exploits the sparsity of updates (typically <5% of parameters change significantly between versions):

$$\text{Size}(\text{Compress}(\Delta\theta)) \approx 0.03 \cdot \text{Size}(\theta) \tag{12}$$

This enables model updates over low-bandwidth connections (satellite phones, intermittent cellular) that would be impractical for full model downloads.

7.3 Battery and Thermal Management

Field deployment requires careful power management. ZooMobile implements an adaptive computation budget:

$$\text{Budget}(t) = f(\text{Battery}(t), \text{Temperature}(t), \text{Priority}(t)) \tag{13}$$

Under low battery or high temperature conditions, the system reduces computation by:

1. Skipping super-resolution preprocessing (saves 15ms per inference).
2. Reducing multi-scale inference from 3 crops to 1 (saves 25ms).
3. Increasing the confidence threshold to avoid uncertain identifications.
4. Caching recent identifications for repeat encounters.

8 Field Deployment

8.1 Deployment Sites

We deployed ZooMobile across 8 national parks in 5 countries over 6 months (July 2025 – January 2026):

Table 7: Field deployment sites and statistics.

Site	Country	Rangers	Target Species	IDs Made
Serengeti NP	Tanzania	24	847	4,210
Kaziranga NP	India	18	612	3,470
Yasuní NP	Ecuador	14	1,234	2,890
Taman Negara NP	Malaysia	16	978	2,640
Kruger NP	South Africa	22	724	3,180
Manu NP	Peru	12	1,087	2,340
Chitwan NP	Nepal	20	543	2,710
Cat Tien NP	Vietnam	16	689	1,960
Total		142		23,400

8.2 Field Accuracy

Of 23,400 identifications, 19,818 (84.7%) were verified as correct by expert review. Accuracy varied by taxonomic group:

Table 8: Field accuracy by taxonomic group.

Group	IDs	Correct (%)	Top-3 (%)	Declined (%)
Large mammals	5,420	91.2	97.4	3.1
Birds	6,180	84.3	93.8	8.4
Reptiles	3,240	86.7	94.1	6.2
Amphibians	2,140	79.4	90.3	11.7
Insects	3,870	78.2	89.4	14.3
Plants	2,550	87.1	95.2	5.8

The "Declined" column shows the percentage of identifications where the model's confidence was below τ_{low} and it provided guidance rather than an identification. Higher decline rates for insects and amphibians reflect genuine difficulty (small body size, variable coloration) rather than model failure.

8.3 Conservation Impact

During the deployment, ZooMobile directly contributed to conservation actions:

- **Anti-poaching:** 47 patrol decisions were informed by species identifications, including 12 instances where identification of critically endangered species triggered immediate alert protocols.
- **Human-wildlife conflict:** 12 interventions were supported by rapid species identification, enabling appropriate response protocols (e.g., distinguishing between venomous and non-venomous snake species encountered in villages).
- **Biodiversity surveys:** Rangers contributed 8,740 verified occurrence records to the Zoo Data Commons, representing a 340% increase in data collection rate compared to manual survey methods.
- **Invasive species detection:** 23 observations of invasive species outside their known range were flagged, triggering early-response protocols at 4 sites.

8.4 User Experience

Ranger satisfaction was assessed through structured interviews at the end of the deployment:

Table 9: Ranger satisfaction survey results (5-point Likert scale).

Dimension	Mean Score
Ease of use	4.3
Identification accuracy (perceived)	3.9
Response speed	4.6
Battery life impact	3.4
Usefulness for patrol decisions	4.1
Overall satisfaction	4.2

Battery life impact received the lowest score. Post-deployment analysis showed that intensive use (50+ identifications per day) reduced battery life by approximately 30%. The adaptive computation budget helped but did not fully address the concern for multi-day patrols without charging infrastructure.

9 Discussion

9.1 Accuracy-Size Tradeoff

ZooMobile demonstrates that the accuracy-size tradeoff for species recognition is more favorable than previously assumed, particularly when domain-specific knowledge (taxonomy) informs the compression process. The 4.5% accuracy gap between ZooMobile (86.8%) and the cloud baseline (91.8%) must be evaluated against the practical alternative: no identification at all when connectivity is unavailable. For 73% of identifications in our deployment, the only alternative to ZooMobile was visual identification by rangers, which averaged 67% accuracy in controlled tests.

9.2 Limitations

Several limitations qualify our results. First, ZooBench-Field, while more realistic than existing benchmarks, was collected at sites with established monitoring programs; truly remote and undersampled regions may present additional challenges. Second, our evaluation focuses on visual species recognition; acoustic identification (bird calls, frog calls) requires a separate pipeline not addressed here. Third, the model update protocol assumes periodic connectivity; sites with zero connectivity for extended periods (months) cannot receive model improvements.

9.3 Ethical Considerations

Deploying species recognition technology raises concerns about misuse for poaching or illegal trade. ZooMobile mitigates these risks by withholding precise location data for IUCN-listed species, requiring ranger authentication for sensitive species information, and implementing audit logging of all identifications with location data accessible only to authorized conservation authorities.

10 Conclusion

We have presented ZooMobile, a framework for deploying species recognition on mobile devices in conservation field settings. Taxonomic Knowledge Distillation compresses models by 94% while preserving 95% of accuracy. Species-specific mixed-precision quantization achieves sub-50ms inference on mid-range smartphones. Condition-Aware Inference adapts to degraded field conditions, recovering up to 5.3 percentage points of accuracy under low-light conditions.

A 6-month field deployment across 8 national parks with 142 rangers demonstrated 84.7% field-verified accuracy, enabling 23,400 real-time species identifications and directly supporting 47 anti-poaching decisions and 12 human-wildlife conflict interventions. ZooMobile closes the deployment gap between laboratory-grade species recognition and the realities of conservation fieldwork.

Future work will extend ZooMobile to acoustic identification, develop continual learning mechanisms for adapting to new species encountered in the field, and integrate with ZooEd for in-field conservation training.

References

Ahumada, J. A., Fegraus, E., Birch, T., et al. (2020). Wildlife Insights: A platform to maximize the potential of camera trap and other passive sensor wildlife data for the planet. *Environmental Conservation*, 47(1):1–6.

Beery, S., Van Horn, G., and Perona, P. (2018). Recognition in Terra Incognita. *Proceedings of ECCV*, pages 456–473.

Berger-Wolf, T. Y., Rubenstein, D. I., and Stewart, C. V. (2017). Wildbook: Crowdsourcing, computer vision, and data science for conservation. *arXiv preprint arXiv:1710.08880.*

Dosovitskiy, A., Beyer, L., and Kolesnikov, A. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *Proceedings of ICLR.*

Han, S., Pool, J., Tran, J., and Dally, W. J. (2015). Learning both weights and connections for efficient neural networks. *Advances in Neural Information Processing Systems*, 28.

Hinton, G., Vinyals, O., and Dean, J. (2015). Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531.*

Howard, A., Sandler, M., and Chen, B. (2019). Searching for MobileNetV3. *Proceedings of ICCV*, pages 1314–1324.

Jacob, B., Kligys, S., and Chen, B. (2018). Quantization and training of neural networks for efficient integer-arithmetic-only inference. *Proceedings of CVPR*, pages 2704–2713.

Liu, Z., Mao, H., Wu, C. Y., et al. (2022). A ConvNet for the 2020s. *Proceedings of CVPR*, pages 11976–11986.

Picek, L., Sulc, M., Matas, J., et al. (2022). Overcoming practical issues of deep active learning with species identification. *Ecological Informatics*, 70:101714.

Schneider, S., Taylor, G. W., and Kremer, S. C. (2020). Three critical factors affecting automated image species recognition. *Ecology and Evolution*, 10(7):3503–3517.

Stevens, S., Wu, J., and Thompson, M. J. (2024). BioCLIP: A vision foundation model for the tree of life. *Proceedings of CVPR*, pages 19412–19424.

Tan, M. and Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *Proceedings of ICML*, pages 6105–6114.

Terry, J. C. D., Roy, H. E., and August, T. A. (2020). Thinking like a naturalist: Enhancing computer vision of fine-grained classification. *Methods in Ecology and Evolution*, 11(11):1360–1374.

Touvron, H., Cord, M., Douze, M., et al. (2021). Training data-efficient image transformers and distillation through attention. *Proceedings of ICML*, pages 10347–10357.

Tuia, D., Kellenberger, B., Beery, S., et al. (2022). Perspectives in machine learning for wildlife conservation. *Nature Communications*, 13:792.

Van Horn, G., Mac Aodha, O., Song, Y., et al. (2018). The iNaturalist species classification and detection dataset. *Proceedings of CVPR*, pages 8769–8778.

Van Horn, G., Cole, E., and Beery, S. (2021). Benchmarking representation learning for natural world image collections. *Proceedings of CVPR*, pages 12884–12893.

Wang, Q., Fink, O., Van Gool, L., and Dai, D. (2022). Continual test-time domain adaptation. *Proceedings of CVPR*, pages 7201–7211.

Wood, C., Sullivan, B., Iliff, M., et al. (2011). eBird: Engaging birders in science and conservation. *PLOS Biology*, 9(12):e1001220.

Yu, W., Luo, M., Zhou, P., et al. (2022). MetaFormer is actually what you need for vision. *Proceedings of CVPR*, pages 10819–10829.

Zhou, K., Liu, Z., Qiao, Y., et al. (2022). Domain generalization: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4):4396–4415.

Antje Worring, Zach Kelling (2025a). Decentralized Semantic Optimization: Collaborative knowledge curation on distributed networks. *Zoo Improvement Proposal ZIP-001*.

Antje Worring, Zach Kelling (2025b). Zoo Network Architecture: A decentralized infrastructure for open science. *Zoo Technical Report ZTR-001*.

Antje Worring, Zach Kelling (2026). Zoo Data Commons: A decentralized repository for open biodiversity datasets. *Zoo Improvement Proposal ZIP-009*.